Sahaj Bhadja

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Summary

Software engineer with 2+ years building **test automation frameworks**, **distributed system tooling**, and **hardware-integrated validation pipelines** using Go/Python/C++ for high-reliability engineering environments.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*
- Built **modular automation services** for 500+ users, reducing workflow latency from **1.2s to 350ms** while improving reliability and diagnostic visibility for lab-style validation tasks.
- Implemented **fault-tolerant execution pipelines** for 5K+ monthly runs with validation and rollback safeguards, improving test stability and post-deployment confidence.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*
- Developed **full-stack validation features** handling 10K+ daily requests, translating requirements into reusable APIs/components and measurable quality controls for release readiness.
- Improved **CI/CD and regression automation**, reducing turnaround by **60%** while preserving production quality in high-iteration environments.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*
- Built **high-volume experiment pipelines** for 100K+ trajectories, enabling reproducible validation and rapid investigation of complex software/hardware behavior interactions.
- Produced **debugging diagnostics and technical reports** that helped cross-functional teams identify root causes and prioritize corrective improvements.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*
- Engineered **distributed backend services** with Python/Java and cloud integrations, improving response time by **30%** across data-intensive operations.
- Maintained **20+ automated test suites** with Jenkins, PyTest, and Mockito on Linux/Windows, increasing pipeline efficiency by **68%**.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*
- Implemented **MERN + WebSocket modules** for eight workflows and 2M+ synchronized records, improving response consistency and real-time system behavior.
- Automated **containerized deployment/testing** with Docker and Jenkins, reducing release time by **50%** and improving rollout reliability.

Skills

Languages: **Golang**, **C++**, **Python**, Java, SQL, Bash
Systems & Automation: distributed services, resource-aware coding, regression test orchestration, lab/production validation
Hardware Integration: microcontroller concepts, CAN/SPI/I2C protocol familiarity, test hardware bring-up support
Infrastructure: Git/GitHub, CI/CD pipelines, Linux/UNIX environments, Docker, cloud deployment workflows
Engineering: root-cause debugging, first-principles problem solving, cross-functional collaboration, code reviews

Projects

Automated Power Test Framework | *Golang, Python, CAN, PostgreSQL*
- Designed a **scalable test automation framework** for hardware validation and regression execution, improving test cycle consistency and reducing manual setup effort across environments.

CPU Scheduling Simulator Live | *C++, React.js, JavaScript*
- Implemented **FCFS, SJF, SRT, RR, and MLFQ algorithms** with runtime metrics and visualization, reducing debugging time by **20%** through repeatable performance analysis.

Linguistix Code | *Python, AWS, Docker, CI/CD*
- Built a **containerized REST pipeline** with automated CI/CD validation, improving operational reliability and sustaining production-grade processing workloads.